Power BI Dashboard Spec - Shipping Route Efficiency

Build this against the `shipments` table described in `schema.md`.

KPI cards, top row

Five cards:

- Total Shipments: `COUNTROWS(shipments)`
- Total Sales: the Total Sales measure below
- Total Orders: `DISTINCTCOUNT(shipments[Order_ID])`
- Avg Order Value: the Avg Order Value measure below
- Avg Lead Time: the Avg Lead Time measure below

Charts

Lead Time Trend: line chart, Order_Month on the x-axis, Avg Lead Time on the y-axis.

Route Performance Leaderboard: horizontal clustered bar chart. Avg Lead Time on the x-axis, Route on the y-axis (use a filter to show the top and bottom 10), and color by Factory. This is the chart that most often gets built wrong, people leave the Factory dimension out entirely and it turns into a generic state ranking instead of a route ranking.

Ship Mode Comparison: clustered bar chart, Ship Mode on the x-axis, with two measures on it, Avg Lead Time and Total Sales. You'll need a secondary axis for this since the two measures are on totally different scales.

Division Sales Share: donut chart, Total Sales as the value, Division as the legend.

Geographic Shipping Map: filled map, State_Province as the location, colored by Avg Lead Time. Remember to set the Data Category on State_Province first or this visual won't render properly.

Region Volume vs. Lead Time: scatter chart, shipment count on the x-axis, Avg Lead Time on the y-axis, colored by Region.

Layout wise, I'd put the KPI row and the trend line at the top, put the map front and center since it's the visual anchor of the whole thing, put the route leaderboard and ship mode comparison side by side

underneath, and tuck the donut into a corner as a smaller supporting chart.

DAX measures

```
Total Sales = SUM(shipments[Sales])

Avg Order Value =
DIVIDE(
    [Total Sales],
    DISTINCTCOUNT(shipments[Order_ID])
)

Avg Lead Time =
AVERAGE(shipments[Shipping_Lead_Time_Days])
```

One more that's useful for the donut chart tooltip:

```
% Sales Share by Division =
DIVIDE(
    [Total Sales],
    CALCULATE([Total Sales],
ALL(shipments[Division]))
)
```

Slicers

- Date range on Order_Date
- Region or State selector

- Ship Mode filter (button slicer works well here)
- Division filter
- A numeric slider on Shipping_Lead_Time_Days so you can isolate slow shipments interactively

Drill-through

Worth setting up a drill-through page from the route leaderboard down to an order-level detail page, filtered to whatever route got clicked, showing Order_ID, Product_Name, Ship_Mode, and Shipping_Lead_Time_Days. Gives you order-level detail without cluttering the main dashboard.

Things that commonly go wrong

These are actual mistakes I caught when checking a first draft of this dashboard against the spec, worth double-checking for before calling it done:

1. The map ends up as a regular bar chart labeled something like "map proxy" instead of an actual filled map. Almost always because the Data Category on State_Province never got set.
2. The route leaderboard drops the Factory dimension and just shows city and state, which defeats the point since the whole idea is comparing factory-to-state routes, not just states on their own.

3. Only the top 10 fastest routes get shown, the bottom 10 slowest get left out. Both halves matter, the slow ones are where you'd actually want to act.
4. The ship mode chart only shows lead time and forgets the sales series, which needs that secondary axis to display properly.